



Welcome! 😊

## Introduction to Software Engineering

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# **A Nice Quote**

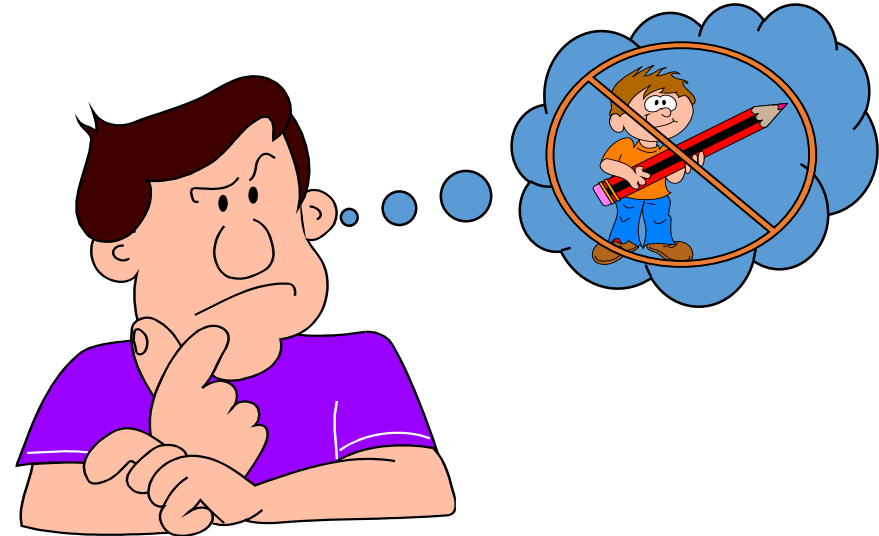
**W**ant to get something in life

**A**lways think positive

**Y**ou will definitely get the thing you want

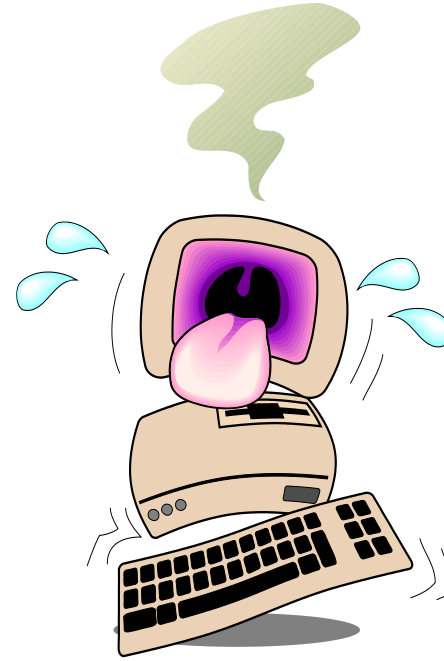
# Five Tips to Success

- Work Hard
- Try More exercises and more practice
- Do the Labs and assignments by yourself

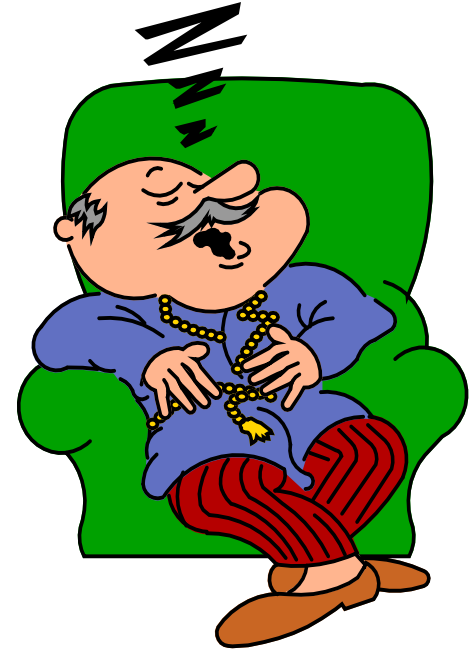


# Five Tips

- **B**e patient with the Machine



- If you really need that, **do it quietly**



Together  
WE  
Achieve  
More





The  
expert in  
anything  
was  
once a  
beginner.

**DO IT NOW.**

**SOMETIMES  
'LATER'  
BECOMES  
'NEVER'**

# Course Objective

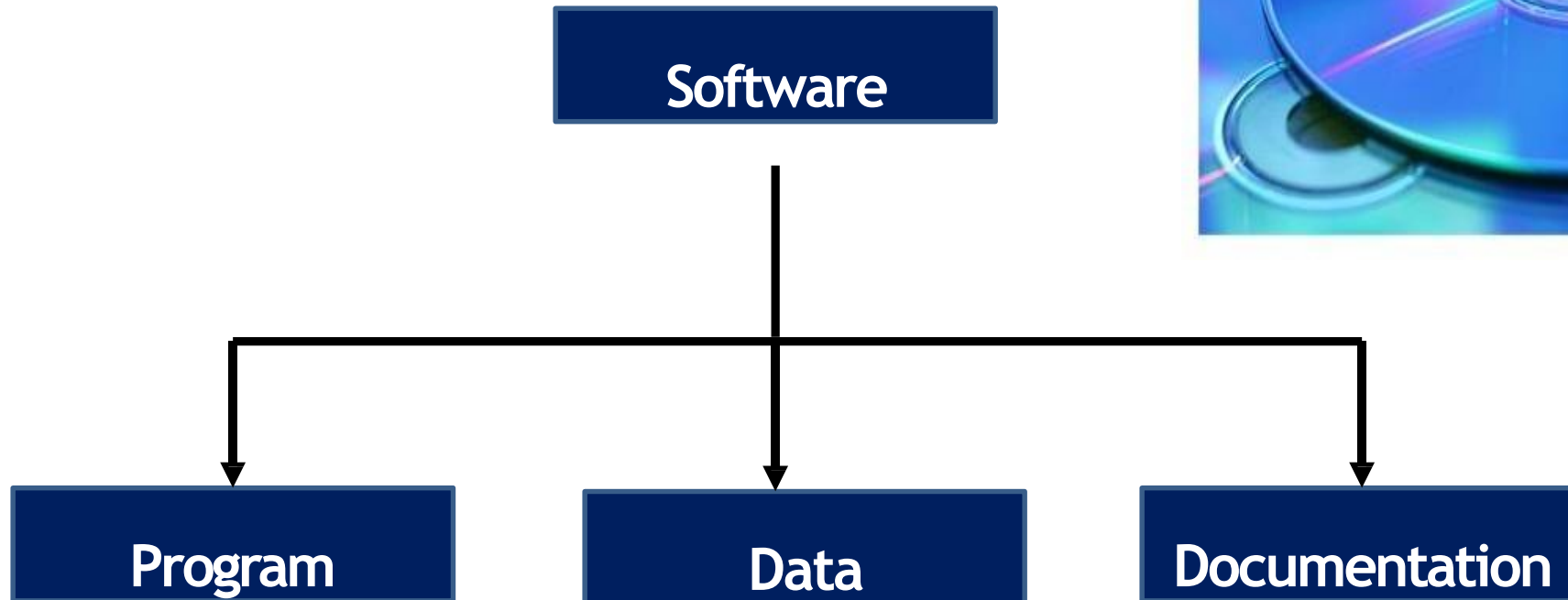
- Upon successful completion of this course the student will be able to apply software engineering techniques effectively to produce a successful software product.
- They will be able to structure the software production process, **analyze the requirements** for a software system, produce a **software design** and **architecture** from requirements.
- The course should also introduce the students to software **project management** and planning issues and various software **validation** and **verification** techniques.



What is Software?

# Software

*Computer instructions or data*



# What is Engineering?

*“The **process** of productive use of **scientific knowledge** is called Engineering.”*



# What is Software Engineering?

*Application of systematic, disciplined, quantifiable approach to development, operation and maintenance of software. (IEEE-Std.)*



# Difference b/w CS and SE

*“ This is the process of **utilizing** our knowledge of **computer science** in effective production of software systems.”*

Software can have huge impact  
in **any aspect** of our society



# Where can we find software?



# Some popular ones...

**facebook**

☐ Remember Me

[Forgot your password?](#)

Email

Password

Login

**Facebook helps you connect and share with the people in your life.**

**Sign Up**

It's free and anyone can join

Full Name:

Your Email:

New Password:

I am:

Select Sex: ▼

Birthday:

Month: ▼

Day: ▼

Year: ▼

Why do I need to provide this?

Sign Up

[To create a page for a celebrity, band or business, click here.](#)

# Some popular ones...



Google Search

I'm Feeling Lucky

Google.com.pk offered in: اردو

And even in...



# Conclusion

Software is almost everywhere!!!

# Problems in software development

- Common issues
  - The final software does not fulfill the needs of the customer
  - Hard to **extend and improve**: if you want to add a functionality later its mission impossible
  - Bad **documentation**
  - Bad quality: **frequent errors**, hard to use, ...
  - More time and costs than expected





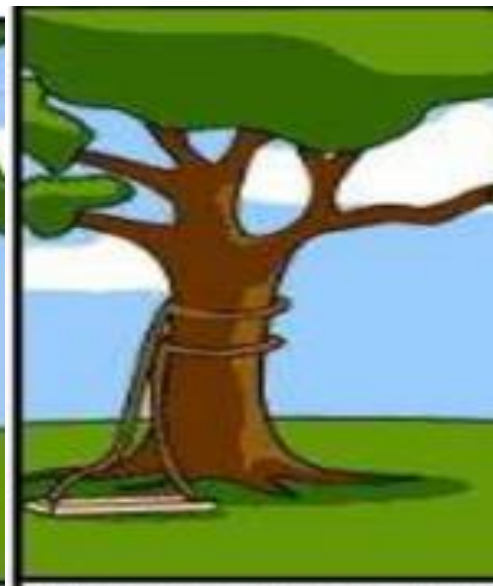
How the customer explained it



How the Project Leader understood it



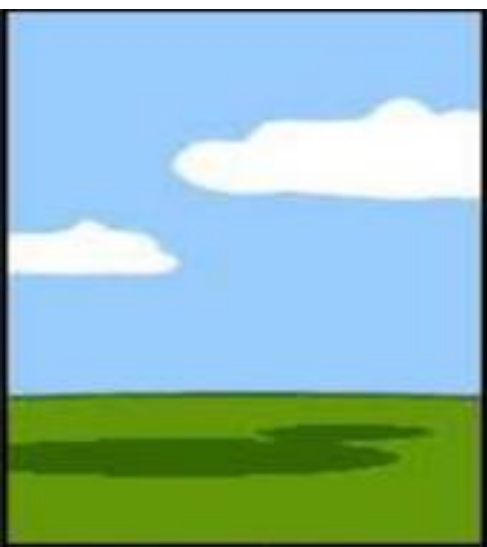
How the Analyst designed it



How the Programme wrote it



How the Business Consultant described it



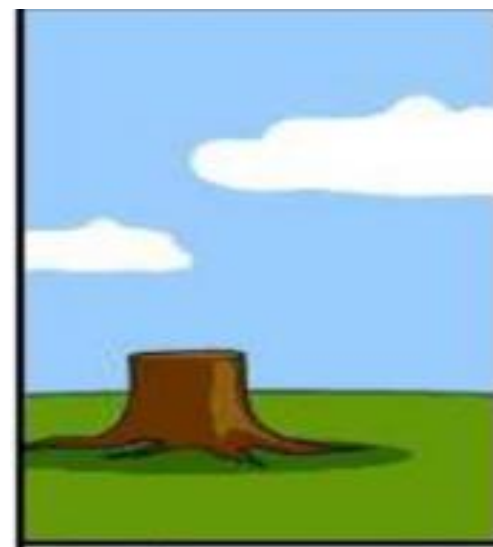
How the project was documented



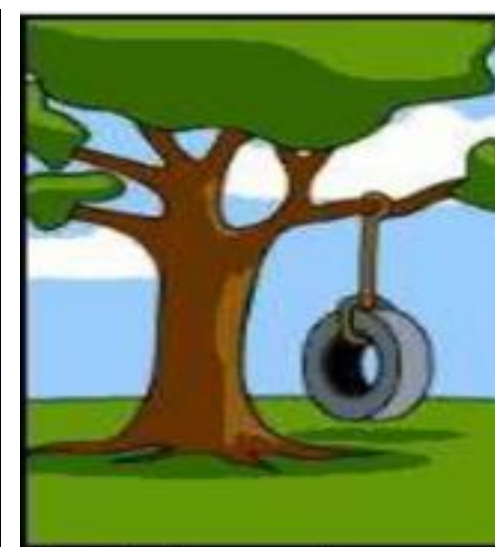
What operations installed



How the customer was billed



How it was supported



What the customer really needed



But

That never  
happens?

Right?

Wrong!

# Ariane 5 Flight 501

- Cause: design errors in the software
- <http://www.youtube.com/watch?v=kYUrqdUyEpl>

# Conclusion

- Programming is NOT enough!

*It is not enough to do your best:*

*you must*

*Know what to do, and THEN do your best.*

W. Edwards Deming

# And Since...

➤ *A **clever** person **Solves** a problem.*

➤ *A **wise** person **Avoids** it.*

- Albert Einstein

Solution?





# Well-Engineered Software

- Reliable
- Good User Interface
- Acceptable Performance
- Good Quality
- Cost Effective

# Software Engineering

1) *The application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is the application of engineering to software.*

*-- IEEE Definition*

# Software Engineering

- Software engineering is about the creation of **large pieces of software** that consist of **thousands of lines of code** and involve **many person months of human effort**.

**(Douglas Bell)**

# Software Engineering

- One of the **attractions** of software engineering is that there is **no one single best method** for doing it, but instead a whole variety of different approaches.
- **Consequently**, the software engineer needs a **knowledge** of many **different techniques** and tools.

# What does a Software Engineer do?

Software engineers apply the **principles** and **techniques** of **computer science, engineering**, and **mathematical analysis** to the **design, development, testing**, and **evaluation** of the software and the systems that enable computers to perform their many applications.

- Software engineers must possess **programming skills**, but are often more concerned with **developing algorithms** and **analyzing and solving programming problems** than with actually writing code.